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Solving Multiple Discretization Portfolio Optimization Problem with Quantum-Classical Hybrid Algorithms

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Abstract
Portfolio optimization (PO) plays a central role in finance, encompassing both resource allocation and risk management. While numerous algorithms have been developed for continuous-variable PO, the inclusion of large, indivisible assets such as real estate transforms the task into a combinatorial optimization problem, which is NP-hard for classical computers. This study explores using quantum-classical hybrid algorithms for multi-discretized portfolio optimization, addressing the needs of real-world applications. We propose a flexible theoretical model that can solve PO problems with various types of discrete assets and constraints, which is compatible with both quantum and classical computing paradigms. Then, we use the D-Wave quantum processor and classical solver to determine optimal investment strategies. Experimental results indicate that the hybrid quantum algorithm outperforms both classical computing methods and quantum annealing algorithms in terms of return and risk management. This research enhances quantum computing applications in finance and provides insights into future algorithm design.

Keywords Portfolio optimization · Discrete combinatorial optimization problem · Quantum computing · Quantum-classical hybrid algorithms

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1 Introduction

Portfolio optimization (PO) is a critical financial task that focuses on finding the optimal balance between risk and return, with wide applications across various financial sectors. This process often considers multiple constraints, such as risk thresholds, investment proportions, diversification requirements, and regulatory compliance (Lewis, 1988). The Markowitz model provides a framework for quantifying the trade-off between risk and return (Markowitz, 1952). Typically, this model is effective when dealing with continuous variables. However, in real-world scenarios, the independent variables consist of both continuous and discrete types. For example, decisions such as whether to purchase a real estate property or a high-value NFT are naturally modeled as binary variables, while the allocation of indivisible high-value homogeneous goods (i.e., aircraft, servers, or heavy machinery) is best represented using integer variables. These variables convert the optimization problem into a Mixed-Integer Optimization problem or Discrete Optimization problem which is NP-Hard, thus challenging to solve classically (Vesselinova et al., 2020). Consequently, there is a pressing need for advanced algorithms capable of effectively solving these complex Combinatorial Optimization Problems (COPs), especially in the field of financial computing (Orús et al., 2019; Lopez de Prado, 2015).

Given the complexity of combinatorial optimization algorithms, developing and refining heuristic algorithms has emerged as a significant research focus in this field. Some notable heuristic algorithms have been widely applied in portfolio optimization, option pricing, risk prediction and some other financial applications (Doğan et al., 2024; Cura, 2021; He et al., 2023; Shinzato, 2018; De Moraes & Coelho, 2022). These include algorithms such as Tabu Search (TS) (Glover, 1986, 1990), Simulated Annealing(SA) (Kirkpatrick et al., 1983), Steepest Descent(SD) (Bottou, 1998; Ruder, 2016), and Genetic Algorithms (Goldberg et al., 1989). Particularly, some nature-inspired algorithms have also demonstrated promising efficacy in addressing some specific optimization problems, such as firefly algorithm (Yang, 2009; Abedi & Gharehchopogh, 2020) and RIME optimization algorithm (Abdel-Salam et al., 2024). With the rapid development of machine learning technologies, some machine learning methods have gradually been used in portfolio optimization and other financial computing scenarios (Eraña-Díaz et al., 2020; Chen, 2021; Orra et al., 2025).

Quantum computing (QC), as a new computational paradigm, offers theoretically grounded advantages in solving prime factorization problem, a NP-hard problem (Shor, 1994). Additionally, QC exhibits significant advantages in addressing Combinatorial Optimization Problems (COPs), which involve discrete variables (Nielsen & Chuang, 2011; Brandhofer et al., 2022; Lu & Yang, 2024; Woerner & Egger, 2019; Orús et al., 2019). Recent advances in quantum hardware—including increases in qubit count (Guo et al., 2024), gate fidelity (Ryan-Anderson et al., 2024), and quantum volume (Jurcevic et al., 2021; Baldwin et al., 2022)—have significantly improved the feasibility of applying quantum algorithms to real-world financial problems (Sofge, 2008; Orús et al., 2019; Lu & Yang, 2024; Buonaiuto et al., 2023). For instance, IBM demonstrated a quantum algorithm based on amplitude estimation for financial risk analysis that achieved faster convergence than classical Monte Carlo simulations (Woerner & Egger, 2019). D-Wave's superconducting quantum annealer

has been employed to solve discrete-time portfolio optimization problems, outperforming classical methods such as GEKKO (Beal, 2021) and quantum-inspired tensor network optimizers (Orús, 2014, 2019; Mugel et al., 2022) in terms of computational speed. Additional applications of quantum annealing include addressing multi-period portfolio optimization (Rosenberg et al., 2015), asset correlation detection (Kalra et al., 2018), and minimum holding period constraints in portfolio rebalancing (Mugel et al., 2021). These developments demonstrate that even in the current Noisy Intermediate-Scale Quantum (NISQ) era, quantum computing is already being effectively applied to practical financial optimization tasks.

In previous work, quantum algorithms were limited to handling portfolio optimization problems for a single asset type. Our model incorporates quantitative constraints by including assets that must be acquired in indivisible units—such as aircraft, servers, or heavy machinery—aligning closely with real-world investment practices. It also accommodates assets with multiple discrete valuations, such as real estate, art, and private equity, thereby capturing the complexity of modern investment decisions. Consequently, our model expands the applicability of quantum algorithms in solving portfolio optimization problems to a broader range of asset types.

We then implemented this model by using the D-Wave quantum annealer alongside various classical solvers, conducting backtesting on two datasets. Results show that the quantum-classical hybrid approach delivers superior performance in terms of return and risk trade-offs. Furthermore, we analyzed the impact of increased qubit counts on solution quality, demonstrating further potential for performance gains. This work establishes a practical framework for applying quantum computing in finance and outlines avenues for future improvement.

This paper is structured as follows: Firstly, we introduce our proposed theoretical model aims at solving multiple discretization portfolio optimization problems. In the Next, we introduce how to introduce linear constraints in the model and transform it into a QUBO problem. Then, we outline the computational methods and datasets containing multiple discretization assets. Subsequently, we execute this model on D-Wave's processor and compare the results with various classical algorithms for solving PO problems. In the following section, the results are shown and the algorithmic complexity and effectiveness of these schemes are discussed. Finally, conclusions and future works are outlined.

2 Mathematical Modeling

2.1 QUBO Interpretation of Multiple Discretization PO Problem

This section elucidates the construction of a discrete portfolio optimization model suitable for both quantum and classical computation. The input data include historical closing prices of selected assets, from which we compute return rates and the covariance matrix to characterize expected performance and risk.

We formulate the problem using a QUBO-based framework to represent portfolio optimization with multiple discrete decision variables. Our approach builds on the classical Markowitz mean-variance model (Markowitz, 1952), which balances

expected return and risk via a risk preference parameter, λ. For a portfolio with up to N assets, the objective function is given by:

$$\min_{\omega} \left(\lambda \omega^T \Sigma \omega - \omega^T \mu\right) \quad s.t. \sum_{i=1}^{N} \omega_i = 1, \tag{1}$$

where Σ is the covariance matrix, and ω depicts the investment strategy. ω_i denotes the weight assigned to the ith asset within the portfolio strategy. The covariance matrix, Σ, depicts the interrelationships of daily returns among assets. The covariance between assets i and j, σ_{ij}, is calculated as

$$\sigma_{ij} = \frac{1}{N_t - 1} \sum_{t=1}^{N_t} (R_{it} - \mu_i)(R_{jt} - \mu_j), \tag{2}$$

where R_{it} is the daily return on day t for asset i determined by the formula $R_{it} = \frac{P_{it} - P_{i,t-1}}{P_{i,t-1}}$. P_{it} is the closing price of asset i on day t. The expected daily return is $\mu_i = \frac{1}{N_t} \sum_{t=1}^{N_t} R_{it}$, with N_t representing the number of trading days considered in the analysis. The covariance matrix, Σ, encapsulates the joint variability and correlation between the returns of multi-assets across time.

This mathematical model can be precisely solved when ω is a continuous variable, but NP hard for discrete variable. However, this assumption is seldom encountered in reality, especially in the context of non-fungible assets like estates and artworks. In our model, we generally categorize assets into three types:

1. Fungible Continuous (FC): Fungible assets that offer a nearly continuous range of values, such as individual funds. where each asset's price is relatively small in comparison to the total budget B.
2. Fungible Discrete (FD): Fungible assets offering discrete valuation options, like aircraft, servers, or heavy machinery.
3. Non Fungible (NF): Non-fungible assets with a fixed value of either zero or a specific amount V, such as private funds with minimum investment requirements, real estate, and artworks.

Therefore, according to the above classification, we rewrite a new objective function as a function of v_i:

$$\min_{v} \left(\lambda \vec{v}^T \sum \vec{v} - B \vec{\mu}^T \vec{v}\right) \quad \text{subject to} \quad \sum v_i = B, \tag{3}$$

Each asset value v_i is drawn from a domain depending its type and represented using binary variables:

$$
\begin{aligned}
&v_i \in [0, B], && v_i = \sum_{t=0}^{K} 2^{-t} B\, b_i^{(t)} && \text{for } i \in FC, \\
&v_i \in \left\{ np_i \mid 0 \le n \le \left\lfloor \frac{B}{p_i} \right\rfloor \right\}, && v_i = \sum_{t=0}^{\left\lfloor \frac{B}{p_i} \right\rfloor} p_i\, b_i^{(t)} && \text{for } i \in FD, \\
&v_i \in \{0, p_i\}, && v_i = p_i\, b_i^{(0)} && \text{for } i \in NF.
\end{aligned}
\tag{4}
$$

Let x denote the vector of all binary variables $b_i^{(t)}$, which is a linear transformation of v. The QUBO objective becomes:

$$
\min \left(\mathrm{x}^T \tilde{\Sigma} \mathrm{x} - \tilde{\mu}^T \mathrm{x} \right) \quad \text{subject to} \quad \sum \tilde{p}_i x_i = B. \tag{5}
$$

To include the linear constraint in the objective function, we adopt a squared penalty method with penalty term and corresponding penalty coefficient ξ, resulting in the QUBO format:

$$
\min \left(\mathrm{x}^T \tilde{\Sigma} \mathrm{x} - \tilde{\mu}^T \mathrm{x} + \xi \left(\sum \tilde{p}_i x_i - B \right)^2 \right). \tag{6}
$$

2.2 Constraints of the Number of Assets

To enhance portfolio diversification, it is often necessary to constrain the number of selected assets within a specified range [L, U]. Such constraints require the inclusion of auxiliary binary variables y_i, indicating whether asset i is included in our portfolio. These constraints can be mathematically expressed as:

$$
L \leqslant \sum_{i=0}^{i<N} y_i \leqslant U, \quad y_i \leqslant \sum_{k=0}^{k<K} b_i^{(k)} \leqslant K y_i, \quad \text{for} \quad i \in [0, N-1]. \tag{7}
$$

To convert Eq. 7 into equalities, slack variables are introduced along with penalty terms. Specifically, the slack variables, s_L and s_U, are introduced for each constraint respectively. Then, Eq. 7 is transformed into equalities as follows:

$$
\sum_{i=0}^{N-1} y_i - L + s_L = 0, \quad \sum_{i=0}^{N-1} y_i - U - s_U = 0, \tag{8}
$$

$$
y_i - \sum_{k=0}^{K-1} b_i^{(k)} + s_{Li} = 0, \quad \sum_{k=0}^{K-1} b_i^{(k)} - K y_i - s_{Ui} = 0, \quad \text{for} \quad i \in [0, N-1], \tag{9}
$$

where $s_L = \sum_{j=0}^{m} 2^j s_{Lj}$ and $s_U = \sum_{j=0}^{m} 2^j s_{Uj}$ are the binary decomposition of slack variables. Here, *m* is the maximum number of binary bits required, which is determined based on the maximum value that each slack variable might attain. It is essential to note that the coefficients in the above formula including the slack variables are all integers. It enables their representation in a binary format to align with the binary essence of the problem. This approach guarantees that the integrity of slack variables as integer (specifically binary) components is maintained in optimization problems dominated by binary decisions.

The penalty terms for constraints on the number of assets are included in the final objective function by introducing the expression:

$$\begin{aligned} &\xi\left((\sum_{i=0}^{N-1} y_i - L + s_L)^2 + (\sum_{i=0}^{N-1} y_i - U - s_U)^2\right) \\ &+ \sum_{i=0}^{N-1}\left(\xi(y_i - \sum_{k=0}^{K-1} b_i^{(k)} + s_{Li})^2 + \xi(\sum_{k=0}^{K-1} b_i^{(k)} - Ky_i - s_{Ui})^2\right). \end{aligned} \tag{10}$$

By adding these penalty terms Eq. 10 to the original objective function described in Eq. 6, the comprehensive objective function for this optimization problem is formulated. This approach enables the application of optimization techniques to identify solutions that adhere to the initial constraints with maximal accuracy. The parameter ξ is adjusted to maintain a balance between the degree of constraint breaches and other optimization objectives.

3 Methods and Dataset

3.1 Dataset of Preselect Assets

In this paper, we prepare two distinct datasets of financial assets to validate our model's effectiveness in portfolio optimization problems. Dataset-1 only contains publicly tradable stocks from the Shanghai and Shenzhen Stock Exchanges, covering over a four-month period from September 1, 2021, to December 31, 2021. Dataset-2 includes three asset types, private equity, mutual funds, and stocks, with five instances of each asset type. The time window of Dataset-2 ranges from January 1, 2024, to April 18, 2024, encompassing 57 trading days. Stocks are categorized as Fungible Continuous (FC), requiring investments in 100-share increments. Mutual funds are considered Fungible Discrete (FD) due to the lack of similar restrictions. Private equities are labeled Non-Fungible (NF), where interpolation methods are used to calculate the value of assets for ensuring data consistency across asset types. All pre-selected assets were obtained through random selection, in consideration of industry diversity and fundamental assessment of assets. (See Appendix A for more details about data.)

3.2 Computational Methods and Algorithms

In this work, we employ various methods and hardware implementations to solve the portfolio optimization problem, which are shown below:

1. Quantum-Classical hybrid algorithm: Quantum-Classical Hybrid Annealing (QCHA) implemented on D-Wave quantum annealing machine (King et al., 2022; McGeoch, 2020).
2. Quantum algorithm: Quantum Annealing (QA) implemented on D-Wave quantum annealing machine (King et al., 2022; McGeoch, 2020).
3. Classical commercial software: Gurobi (Gurobi, 2018).
4. Classical inspired algorithm: Simulated Annealing (SA) (Kirkpatrick et al., 1983).
5. Classical algorithm: Tabu Search (TS) (Glover, 1986, 1990).
6. Classical algorithm: Steepest Descent (SD) (Bottou, 1998; Ruder, 2016).

QCHA and QA are executed on the D-Wave Advantage system4.1 via the D-Wave cloud and software (DWAVE, 2023). Compared to QA, QCHA leverages the advantages of quantum algorithms without being limited by the number of qubits by decomposing the master problem into several sub-problems with fewer independent variables. These sub-problems are solved by using quantum annealing algorithms, and their optimal solutions are recombined to update the global solution. Consequently, QCHA is capable of solving larger-scale optimization problems while maintaining a solution quality compared to QA.

In this paper, these four classical algorithms are implemented on PC served as benchmarks for quantum algorithms. These classical algorithms represent some of the most common and effective methods for solving optimization problems. Gurobi (Gurobi, 2018) is a widely recognized commercial software package that specializes in solving combinatorial optimization issues. The other three algorithms are similarly effective in solving combinatorial mathematical optimization and are extensively utilized in various fields, including finance (Rubio-García et al., 2024; Karakalidis & Sifaleras, 2016; Mansouri & Moghadam, 2021; Doğan et al., 2024), job scheduling (Ma et al., 2023; Xie et al., 2023; Mahjoubi et al., 2022), path planning (Ait-Saadi et al., 2022; Sheikh-Hosseini & Hashemi, 2022), and machine learning (Araya, 2021; Song & Yang, 2023; Ganie & Dadvandipour, 2023). (See Appendix C for more details of the Computational methods.)

3.3 Evaluation Indicators of Portfolio Optimization

In this work, we use the objective function value—analogous to the expectation value of the cost Hamiltonian in quantum algorithms—to determine convergence. However, relying solely on this metric to evaluate portfolio optimization neglects the risk aspect of portfolio. Instead, we assess performance using Net Asset Value (NAV) (Williams, 1938) to compute daily returns over shorter horizons. We further incorporate standard financial metrics such as the rate of return (RET) (Cartea et al., 2015; Markowits, 1952), the accumulated return (ACC) (Cartea et al., 2015), the

Sharpe ratio (Sharpe, 1966), and the Sortino ratio (Sortino & Van Der Meer, 1991) to provide a more robust and comprehensive evaluation. (See Appendix B for details.)

4 Experimental Results

In this section, we perform the results of portfolio optimization achieved through the implementation of various algorithms and parameter configurations. Our study includes classical algorithms and quantum algorithms running on D-Wave quantum annealing machines. We detail the performance of these algorithms using two datasets, Dataset-1 and Dataset-2, and discuss how well the theoretical models align with practical applications. Then, we present the performance of quantum algorithms when introducing constraints and compare it to classical algorithms. Finally, we discussed the advantages of quantum algorithms in portfolio optimization problems, and the effectiveness of our theoretical model and propose possible future optimizations based on the results from the actual datasets.

4.1 Performance of Multiple Algorithms

Firstly, we compute the covariance matrices and corresponding objective functions of pre-selected assets in both Dataset-1 and Dataset-2, respectively. Secondly, six different algorithms are employed to solve the portfolio optimization problem and obtained the optimal investment strategies. Notably, the Quantum Annealing Algorithm (QA) is not applicable to Dataset-2 due to the hardware limitations of the D-Wave system, as the required number of qubits exceeded its capacity. In the next, we back-test these strategies by the five evaluation indexes. The backtesting rule involves opening positions based on the optimization results from different algorithms at the start of the trading period, with the assumption that no trading occurs during this period.

Figure 1(a) illustrates the daily return, NAV, over the trading period calculated by six different algorithms in Dataset-1. For each stock, we assign five binary dis-

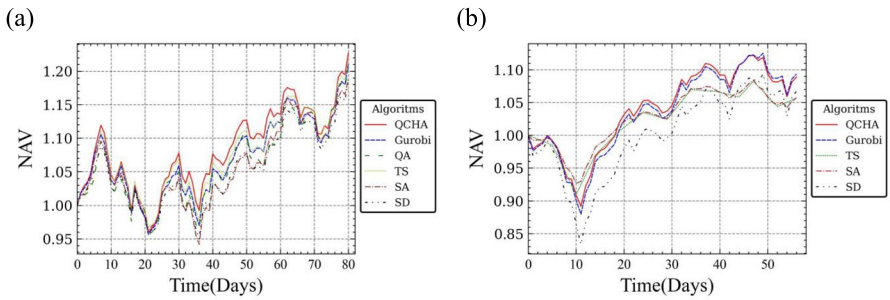

Fig. 1 The daily return of portfolio optimization strategies in Dataset-1 and Dataset-2. (a) For Dataset-1, the daily fluctuation of the NAV is illustrated using curves in different colors, representing six algorithms, QCHA, QA, TS, SA, and SD. In this case, five binary variables are used to discretize the investment proportions for each stock. The x-axis represents trading days, excluding non-trading days. (b) For Dataset-2, five algorithms are used for portfolio optimization, where 128 binary independent variables are assigned to discretize the investment ratios for 15 assets

crete independent variables to represent investment proportions, which is equivalent to using 50 quantum bits in all. The analysis of Fig. 1(a) reveals that while the investment strategies generated by the six algorithms differ, the yield variation trends over time exhibit a considerable degree of consistency. Notably, CHA algorithm, depicted by the solid red line, outperforms the other algorithms consistently over the entire duration, showcasing a superior end-of-period yield assessment. The same calculations were replicated on Dataset-2, as shown in Fig. 1(b). It is evident that both QCHA (solid red line) and Gurobi (dashed blue line) algorithms maintain advantages throughout the entire time period, compared to the other three algorithms. Analysis of these two datasets underscores the superiority of QCHA in backtesting concerning daily NAV. (See Appendix D for specific allocation proportions of investment strategies and more details.)

From the perspective of financial investment practitioners, the values of returns and risks at end-of-period are of primary concern. Figure 2(a1-a3) display (a1)ACC, (a2)Sharpe ratio, and (a3)Sortino ratio obtained by six algorithms at the end of the period respectively. These calculations are based on Dataset-1, utilizing 50 discrete variables. It is evident that the QCHA algorithm exhibits advantages in both ACC and Sharpe ratio. Although it may not exhibit the best performance in Sortino ratio, the difference compared to the best-performing is marginal. This result indicates that QCHA is able to deliver high overall returns with strong risk-adjusted performance, making it an appealing choice for practitioners focused on maximizing utility.

Similarly, we conducted similar experiments in Dataset-2, as depicted in Fig. 2(b1-b3). Notably, Fig. 2(b3) highlights that the strategy produced by the QCHA algorithm excels in the Sortino ratio. While in terms of ACC and Sharpe ratio, QCHA may not be the optimal as indicated in Fig. 2(b1) and (b2). This suggests that QCHA maintains

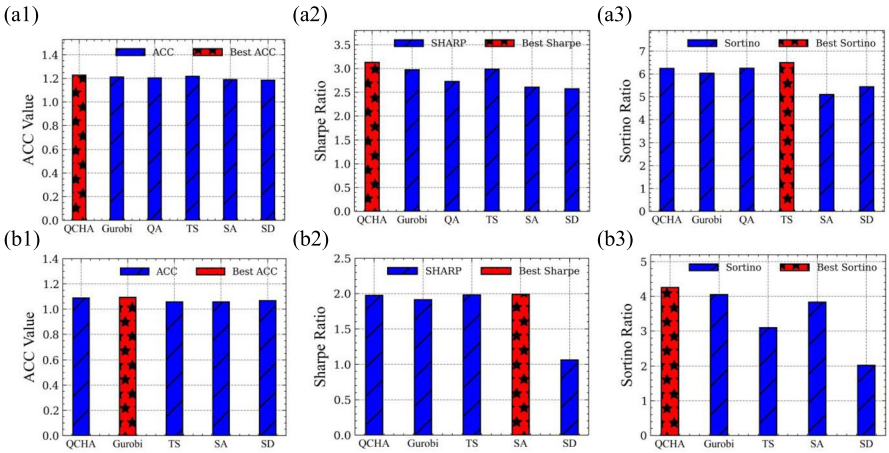

Fig. 2 The end-of-perioid evaluation of portfolio strategies provided by different algorithms based on (a) Dataset-1 and (b) Dataset-2 respectivily. The evaluations are presented by three indexes: (1) ACC values, (2) Sharpe ratio, and (3) Sortino Ratio. The number of discretized independent variables is 50 for Dataset-1 and 128 for Dataset-2. In each set of numerical experiments, the algorithm with the best performance is especially marked as a red box with black stars

robust downside protection while offering competitive returns, thereby providing a conservative yet efficient investment strategy in risk-sensitive environments.

Through the comparison of experiments illustrated in Figs. 1 and 2, QCHA demonstrates the most outstanding performance in both two datasets, showcasing its adeptness in balancing returns and risks. Additionally, the Gurobi algorithm delivers notable results, serving as a benchmark for subsequent experiments.

4.2 Effect of the Number of Qubits on Optimizers

The number of qubits constraints the resolution of Fungible Continuous(FC) assets' proportion in the portfolio. One requires more qubits to encode a high precision float number. On the other hand, more qubits poses challenge to the current quantum hardware and more hard to converge. In this section, we analyze the number of qubits' impact on the financial indicators.

In Fig. 3, we adjust the number of qubits in using QCHA. For Dataset-1, we vary the qubit number from 20 to 50 and evaluate performance via backtesting. The Gurobi algorithm with 50 discrete variables serves as a classical benchmark (red dashed line). The QCHA algorithm with 40 qubits already surpasses Gurobi, underscoring

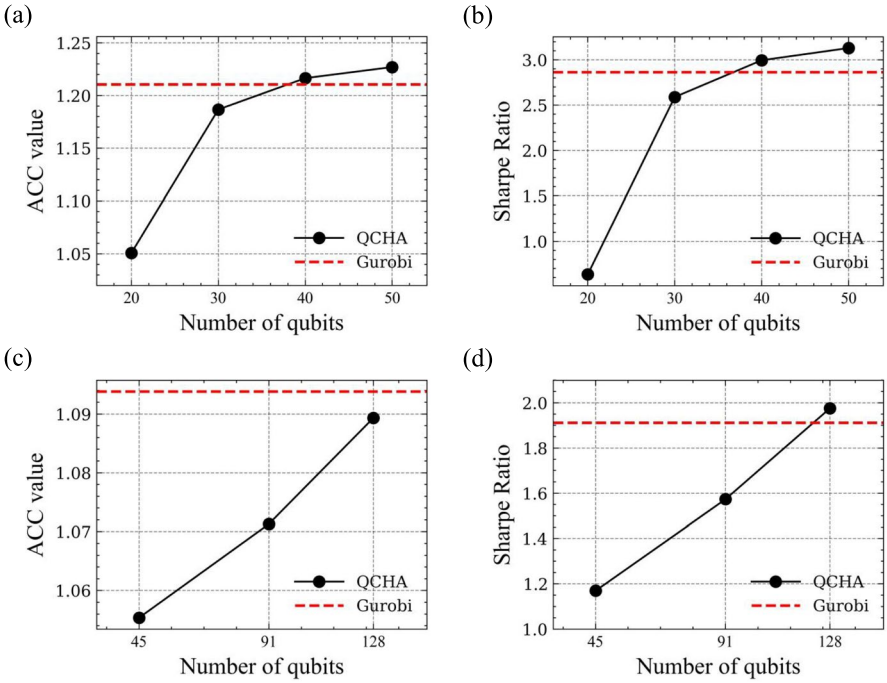

Fig. 3 Portfolio optimization results obtained by the QCHA algorithm with varying numbers of qubits across two datasets. Subfigures (a) and (b) show the ACC values and Sharpe ratios of portfolio schemes solved by QCHA based on Dataset-1, when the number of qubits increases from 20 to 50. The red dashed line represents results from the Gurobi algorithm using 50 discrete variables as a benchmark. Subfigures (c) and (d) present the results using 45, 91, and 128 qubits on Dataset-2, benchmarked against Gurobi with 128 variables

the potential advantage of quantum algorithms. As the number of qubits increases, the precision in investment allocation improves, leading to better outcomes. Financially, this allows for more accurate portfolio weights, enhancing return potential while keeping risk in check.

For Dataset-2, we test QCHA with 45, 91, and 128 qubits, respectivily. Results are shown in subfigures (c) and (d), where performance is evaluated by ACC and Sharpe ratio. With 128 variables, QCHA achieves a higher Sharpe ratio than Gurobi at the same scale. Although its ACC is slightly lower, the trend from 45 to 128 qubits is approximately linear. This indicates steady financial improvement with hardware scaling, supporting finer risk-return tradeoffs.

To summarize, in portfolio optimization, quantum-classical hybrid algorithms already demonstrate superior performance at equivalent problem scales, compared to other four classical solver. With the continued development of quantum hardware and the natural suitability of quantum algorithms for COPs, quantum computing represent a highly promising direction for solviong financial optimization tasks in the future. Quantum computing offer scalable, high-quality solutions for solving multiple discretization portfolio optimization problem.

4.3 Portfolio Optimization with Constraints on the Number of Invested Assets

For solving practical PO problems, incorporating real-world constraints into the theoretical model is a key task to build a robust and efficiency model. For instance, risk-aware investors often diversify across multiple stocks or industries. Imposing minimum or maximum thresholds on the number of investment assets enhances diversification and improves risk management. Our portfolio optimization model integrates constraints on the number of invested assets, as formulated in Eq. 10, using transformations defined in Eqs. 7 and 8. With growing investment universes, incorporating such constraints is crucial for maintaining tractability and relevance.

Figure 4 presents results generated by three solvers—TS, Gurobi, and QCHA—under both constrained and unconstrained settings. For Dataset-1, we assign 4 and 5 binary variables per asset, which equals to 40 and 50 qubits used and corresponds to configurations labeled $40Q$ and $50Q$ on the X-axis. The constraint on the number of invested assets is set to $3 < N < 8$. When this constraint is applied, the total number of discrete variables increases to 87 and 97, respectively.

All four evaluation metrics show a slight decline when constraints are imposed. This can be attributed to the increased complexity of the solution space due to more variables and the structural limitations introduced by the constraints, which can hinder the algorithm's ability to escape local minima. However, the performance drop in QCHA is significantly smaller than TS and Gurobi. This suggests that QCHA maintains high solution quality regardless of constraint inclusion or problem size.

In summary, the introduction of constraints imposes additional challenges on both modeling and algorithmic performance, requiring efficient search strategies and robustness against local optima. Our results demonstrate that the quantum-classical hybrid QCHA algorithm is well-suited for solving constrained portfolio optimization problems in large, discrete solution spaces.

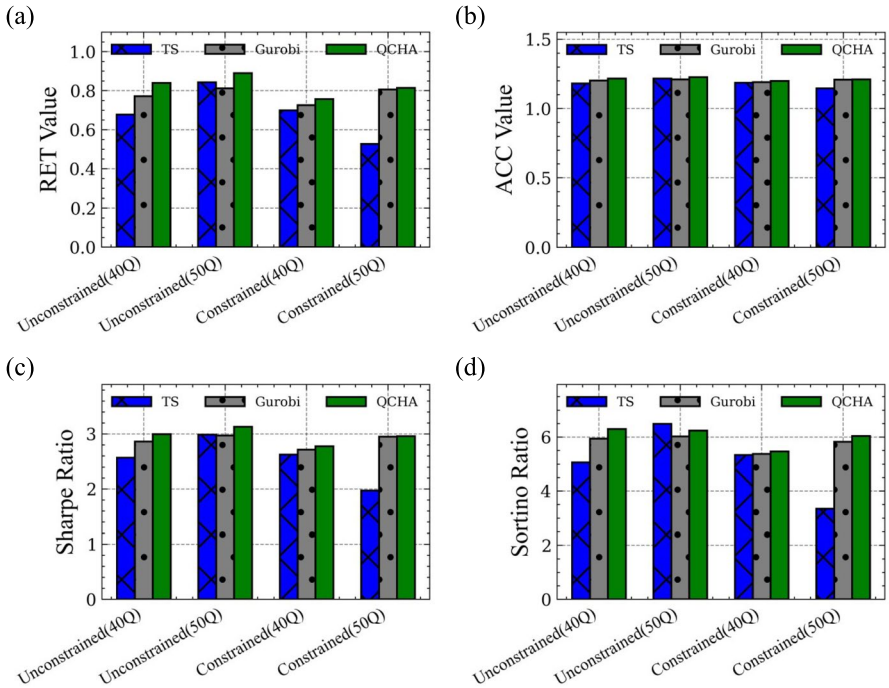

Fig. 4 Portfolio optimization results with and without constraints on the number of invested assets based on Dataset-1. Evaluation metrics include: (a) End-of-period return (RET value), (b) ACC value, (c) Sharpe ratio, and (d) Sortino ratio. Results are generated using three algorithms: QCHA, Gurobi and TS. The number of used discrete variables, equal to qubits in quantum algorithm, is labeled on the X-axis

5 Conclusion and Further Work

In this work, we present a hybrid quantum algorithm designed for multiple-discretization portfolio optimization. This framework allows for the inclusion of non-fungible and high-value indivisible assets, addressing practical challenges often encountered in real-world investment scenarios. The algorithm builds on a QUBO formulation of the Markowitz model and is readily extensible to accommodate various real-world constraints. We evaluate its performance on D-Wave's quantum computing platform and find that, compared to both classical and alternative quantum methods, our approach achieves superior returns while maintaining robust risk resistance. Notably, its advantages become increasingly evident as the number of discrete independent variables (qubits) scales up.

Looking ahead, the integration of real-time market data for dynamic portfolio optimization represents a promising extension. This would require the development of a time-series modeling framework tailored to multiple discretization levels. As quantum hardware continues to evolve, optimizing larger and more complex portfolios becomes feasible. More broadly, the prevalence of binary and integer decision variables across microeconomic models highlights the potential of quantum algorithms in solving combinatorial optimization problems. In order to address larger-scale port-

folio optimization challenges, quantum computing hardware, particularly in terms of gate fidelity and qubit count, requires further enhancement to satisfy the demands of practical applications. Moreover, combining machine learning with quantum computing in this domain offers a compelling avenue for future research and application.

Appendix

A Description to Dataset Selections

In this paper, we prepare two sets of assets, Dataset-1, which includes only normal tradable stocks, and Dataset-2, which comprises private offering funds, public offering funds, and stocks. Both of them exhibit relatively short time windows characterized by short-term unpredictability and volatility. The fluctuations inherent in these short time frames pose significant challenges. This volatility necessitates a more robust strategy, as traditional reliance on broader, long-term trends proves insufficient. These factors highlight the complexity and nuanced nature of formulating short-term financial investment strategies. Consequently, we aim to develop a new algorithm capable of efficiently calculating the optimal solution to mitigate investment risk.

A.1 Dataset-1: Only Stocks

The stocks in Dataset-1 are selected from the Shanghai and Shenzhen stock exchanges for the period spanning September 1, 2021, to December 31, 2021. The dataset comprises 10 stocks and covers a four-month analysis period. Daily closing stock prices are collected from the official websites of the Shanghai and Shenzhen stock exchanges. These two market indices do not exhibit static or monotonic behavior during the designated trading period.

Considering the randomness of the selection of data sampling and the dispersion of asset matching, the selected stocks represent a diverse range of industries, including shipping ports, gaming, power grid equipment, electricity, diversified finance, medical services, and alcoholic beverages. Approximately 50 stocks were chosen from each industry to form a stock pool based on the department's focus. Due to the current limitations in the number of available qubits, sampling methods were employed to select 10 stocks from this pool. Stocks 1 to 10 represent: Rizhao Port (600017.SH), Tianzhou Culture (300148.SZ), Moen Electric (002451.SZ), COSCO Shipping Holdings (601919.SH), and Guangdong Electric Power A (000539.SZ), Xiangyi Rongtong (600830.SH), Tongce Medical (600763.SH), Lianyungang (601008.SH), Kweichow Moutai (600519.SH) and Wuliangye (000858.SZ) respectively.

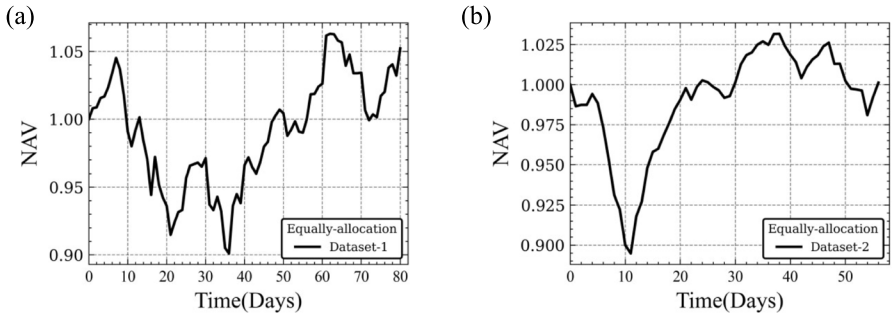

Fig. 5 The Net Asset Value (NAV) over time when the investment amount is equally divided among the pre-selected assets in (a)Dataset-1 and (b)Dataset-2 respectively

As illustrated in Fig. 5(a), we present the daily NAV resulting from an equal investment in 10 stocks from Dataset-1. Backtesting this equal-weighted strategy yields the following end-of-period evaluation metrics: $RET = 0.172$, $ACC = 1.052$, $Sharpe = 0.639$, and $Sortino = 1.245$, with net income close to zero. These results suggest that the selected assets in Dataset-1 display a high degree of randomness and volatility. Compared to the outcomes achieved using various algorithms discussed in the main text, the equal investment strategy appears to be less effective, indicating that the portfolio optimization methods may offer performance improvements over naive diversification in this context.

A.2 Dataset-2: Stocks, Private Equities and Mutual Funds

Dataset-2 comprises 15 financial assets, categorized into five stocks, five public funds, and five private equity funds. The data are sourced from the Wind Database and the stock exchange, covering the period from January 19, 2024, to April 9, 2024. These assets span a broad range of industries, including medical equipment, port operations, cultural entertainment, the low-carbon economy, new energy vehicles, and biomedicine. The method of selecting pre-selected assets is similar to that of Dataset-1, which is random selection from high-quality asset pools across diverse industries.

The selected assets in Dataset-2 are listed as follows:

- **Stocks:**Kweichow Moutai Co., Ltd (600519.SH)
- Jiangxi Guoke Defence Group Co., Ltd (688543.SH)
- NINGBO OCEAN SHIPPING Co., Ltd (601022.SH)
- Well Lead Medical Co., Ltd (603309.SH)
- Shanghai International Port (600018.SH)

- **Public Funds:**CICC Science and Technology Innovation Theme (501080.OF)
- China Merchants CSI Baijiu (161725.OF)
- Yi Fangda CSI Mainland Low Carbon Economy ETF (516070.OF)
- Huitianfu CSI New Energy Automobile Industry C (501058.OF)

- Huaan Guozheng Bio-Pharmaceutical ETF (159508.OF)

- **Private Equity Funds:**Barrow Steady No. 1 (XT1613496.XT)
- Baqisuo Endurance Steady No. 1 (XT1519228.XT)
- Haiwei Quantitative No. 1 (XT1623529.XT)
- Hanxin No. 1 (XT1410420.XT)
- Yitong Growth (XT1510216.XT)Similarly, we conducted a backtest of Dataset-2 under the equal investment strategy. The corresponding results are $RET = 0.005$, $ACC = 1.001$, $Sharpe = -0.159$, and $Sortino = 0.051$. The backtesting indicates that the equal investment strategy has a slight probability of negative returns, approaching 0. This also suggests that dataset-2 exhibits randomness and volatility.

All data were sourced from the public database WIND.These two data selections reflect a real-world situations where assets of discrete choices of buy-in amount are present in our candidate for portfolio optimization. These two datasets highlight the complexity and volatility of formulating short-term trading strategies. Unlike long-term studies, macro trends fail to give an excellent trading strategy. Developing short-term trading strategies with significant fluctuations and randomness poses greater challenges, emphasizing the complexity.

B Introduction to the Evaluation Indicators of Backtest Portfolio Optimization Strategy

The rate of return is a fundamental metric for assessing investment performance, representing the value growth of an investment over a specific period. It is typically expressed as a ratio over time and calculated as follows (Cartea et al., 2015; Jordan et al., 2003):

$$R(t) = \frac{P(t) - P(t-1) + D(t)}{P(t-1)}, \tag{11}$$

where $P(t)$ denotes the end-of-period price at time t, and $D(t)$ represents any dividends or interest received at t. The rate of return $R(t)$ serves as a core indicator for evaluating investment performance.

Net Asset Value (NAV) (Williams, 1938) is a commonly employed measure for assessing the value of an investment portfolio. It is calculated by subtracting total liabilities from total assets. The formula for NAV is:

$$NAV(t) = NAV(t-1) \times (1 + R(t)), \tag{12}$$

where $NAV(t)$ represents the net asset value at time t, and $R(t)$ is the rate of return. In practical applications, NAV is widely used in fund valuation to compute daily returns per share or per fund, making it particularly suitable for evaluating the daily performance of multi-asset portfolios.

The rate of return (RET) (Cartea et al., 2015; Markowits, 1952) standardizes the cumulative return over a specific investment period, N_t, into an equivalent annualized rate. It is calculated using:

$$RET = \left[\prod_{t=1}^{N_t}(1 + R(t))\right]^{\frac{252}{N_t}} - 1, \tag{13}$$

where N_t is the total number of trading days in the investment period. Notably, in this study, RET represents the end-of-period return rather than daily returns (e.g., Fig. 4).

Accumulated return (ACC) (Cartea et al., 2015) reflects the total return of an investment over a period, incorporating the effects of compounding and reinvestment. It is computed as:

$$ACC = \prod_{t=1}^{N_t}(1 + R(t)) - 1, \tag{14}$$

where $R(t)$ is the return at time t. ACC is particularly effective for evaluating long-term investment performance.

The Sharpe ratio, introduced by William Sharpe, is a widely-used metric for assessing risk-adjusted returns (Sharpe, 1966). It compares the excess return—defined as the return above a risk-free rate—to the total investment risk, measured by return volatility:

$$Sharpe = \frac{RET - R_f}{\sigma_{RET}}, \tag{15}$$

where RET is the value from Eq. 13, R_f is the risk-free rate, and σ_{RET} is the standard deviation of returns. A higher Sharpe ratio indicates more favorable risk-adjusted performance. In this analysis, we set $R_f = 0.03$ to reflect a realistic long-term estimate of the risk-free rate in developed markets.

The Sortino ratio refines the Sharpe ratio by focusing solely on downside risk (Sortino & Van Der Meer, 1991). Downside risk, σ_D, is the standard deviation of negative returns and is defined as:

$$\sigma_D = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\min(RET_i, 0))^2}, \tag{16}$$

where N is the number of periods and RET_i is the return for period i. The Sortino ratio is then calculated as:

$$Sortino = \frac{RET - R_f}{\sigma_D}. \tag{17}$$

To emphasize the penalization of losses alone, we adopt a conservative assumption of $R_f = 0$ when computing the Sortino ratio. This convention is commonly used in practice to isolate downside volatility without attributing any baseline reward to risk-free investment.

In this study, NAV is used to monitor real-time portfolio changes and evaluate performance over short time horizons. The four metrics—RET, ACC, Sharpe, and Sortino—are employed to assess portfolio optimization strategies at the end of the investment period. This comprehensive evaluation framework supports more informed and effective investment decision-making.

B.1 Framework of the Quantum–Classical Hybrid Algorithm (QCHA)

The goal of this study is to design an investment strategy that remains effective across a wide spectrum of assets and realistic investment constraints. We cast the multiple-discrete portfolio-selection problem as a quadratic-unconstrained binary optimization (QUBO) task. After collecting the closing-price data of the chosen assets, we compute their return vectors and covariance matrices, then embed the portfolio objective and its constraints into the QUBO matrix Q. Because today's quantum processors offer only a limited number of physical qubits, we adopt a quantum–classical hybrid approach that partitions the large QUBO into smaller sub-problems, each small enough to run accurately on quantum hardware. The overall workflow is summarized in Algorithm 1.

Algorithm 1 HybridQUBOSolve: quantum–classical algorithm for large QUBOs

```
procedure HybridQUBOSolve(Q, max_size, grp_mtd)
    n ← len(Q)
    cur_global ← random binary vector of length n
    best ← cur_global, best_val ← ∞
    while convergence criterion not met do
        grps ← GroupVariables(Q, best, grp_mtd, max_size)
        sel_grps ← SelectSubsets(grps, m, n)
        for all idx ∈ sel_grps do
            subQ ← Q[idx, idx]                                  ▷ extract sub-matrix
            subSol ← QuantumSolve(subQ)
            cur_global[idx] ← subSol
        end for
        cur_global ← LocalSearch(Q, cur_global)
        cur_val ← Evaluate(Q, cur_global)
        if cur_val < best_val then
            best ← cur_global, best_val ← cur_val
        end if
    end while
    return best
end procedure
```

Within each iteration, the *impact index*

$$I_i(\mathrm{x}; Q) = f\big(x_1, \ldots, x_i \oplus 1, \ldots, x_N;\, Q\big) - f(\mathrm{x}; Q), \tag{18}$$

measures the change in objective value when bit x_i is flipped (Booth et al., 2017; Atobe et al., 2021). Variables with similar impact indices are deemed highly correlated and grouped together, which yields sub-problems capturing the most influential interactions. Each sub-matrix subQ is then solved on D-Wave's *Advantage 4.1* annealer via the D-Wave cloud, following the partition-and-iterate strategy of Booth et al. (2017). After the sub-solutions are stitched back into a global vector, a classical local search—such as tabu search or greedy descent—refines the result, updates the incumbent best solution, and the process repeats until convergence. All classical computations were run on a quad-core Intel i5-8265U laptop under 64-bit Windows 11, and all code was written in Python 3.10.13. This hybrid scheme therefore leverages quantum hardware where it is most effective—on tightly scoped, high-fidelity sub-tasks—while retaining the scalability and flexibility of classical optimization for global coordination and convergence.

C Classical Computational Methods

C.1 Introduction to Classical Optimization Solution Algorithms

In this study, four classical algorithms are utilized as benchmarks for evaluating quantum algorithms: Gurobi, Simulated Annealing (SA), Tabu Search (TS), and Steepest Descent (SD). These algorithms are frequently used to address combinatorial mathematical optimization and operations research problems, known for their universality and efficiency. Additionally, they have significant applicability in the financial sector, offering significant practical relevance. Subsequently, we will discuss the fundamental principles and financial application scenarios of these four classical algorithms.

Firstly, the Gurobi optimizer, as a well-known commercial solver, can efficiently handle various combinatorial optimization problems with both discrete and continuous variables. Gurobi's solvers and cloud platform are capable of efficiently solving combinatorial optimization mathematical models. The QUBO-based multi-discretized asset portfolio optimization model proposed in this article is fully compatible with Gurobi. As a non-open-source commercial software, Gurobi integrates various algorithms to solve optimization problems, but its internal algorithmic processes and methods remain proprietary. Nevertheless, available literature indicates that Gurobi demonstrates excellent solver performance and is widely utilized in the field of finance (Karakalidis & Sifaleras, 2016; Min et al., 2023; Pouzada, 2019). In addition, Gurobi solvers are widely used in other field, such as biomedical (Muley, 2021; Yevseyeva et al., 2019), production scheduling (Cortés et al., 2024), power dispatching (Germscheid et al., 2023).

Simulated Annealing (SA) is a probabilistic technique used to approximate the global optimum of a given function. SA is effective in locating the global optimum, particularly when the search space is discrete. In 1983, Kirkpatrick et al. (1983) applied this technique to solve the traveling salesman problem, subsequently coining the term "simulated annealing." Tabu search (TS) is a metaheuristic search method employing local search methods used for mathematical optimization. Tabu search enhances the performance of local search by relaxing its basic rule, which discourages revisiting

previously-visited solutions. It was created by Fred W. Glover in 1986 Glover (1986). Gradient descent is a method for unconstrained mathematical optimization, which is a first-order iterative algorithm for minimizing a differentiable multivariate function. In this work, we utilize the stochastic gradient descent (SGD) algorithm (Bottou, 1998) as a solver for solving portfolio optimization problems, which is also a widely employed tool in contemporary machine learning. The above three classic algorithms for solving optimization problems are also widely applied across various industries, such as finance(Doğan et al., 2024), industrial task optimization, and flight optimization.

C.2 Experimental Results of Classical Algorithms

For classical algorithms used to solve COPs, particularly heuristic approaches, the quality of solutions is highly sensitive to hyperparameter settings and the compatibility between the algorithm and the specific problem instance. Consequently, careful tuning of hyperparameters is crucial for achieving high-quality results and should be tailored to the characteristics of each problem.

In Section 4, we utilize four classical algorithms as benchmarks to evaluate the performance of our proposed quantum algorithm. In the main text, hyperparameters are selected to reflect the optimal performance of each classical solver. This appendix subsection provides a detailed account of classical algorithm performance under varying hyperparameter configurations.

Figures 6 and 7 display the results of solving a portfolio optimization problem on Dataset-1 and Dataset-2, respectively, using classical algorithms under different hyperparameter configurations. Each scatter point in the figures corresponds to an outcome under a specific parameter setting. Due to visualization constraints, only 2–3 hyperparameters are shown. Based on the initial scatter results, we performed further fine-tuning to identify the best-performing solution, indicated by the red triangle. These optimal solutions are reported in the main text as representative benchmarks for each algorithm. For comparison, the QCHA algorithm's results are overlaid as a light-blue plane in each figure. The legend also summarizes the best metric values achieved by both QCHA and the classical solvers.

The alphabetical labels (a)–(d) correspond to the four classical solvers: (a) Gurobi, (b) TS, (c) SA, and (d) SD, with each solver displayed in a separate row of the figure. For the Gurobi solver, we tune two hyperparameters: Time_limit, which specifies the maximum runtime in seconds (higher values allow more time to prove optimality but may increase computation costs), and cuts, which controls the aggressiveness of cutting-plane strategies used to tighten the LP relaxation. Disabling cuts can yield faster approximate solutions, while enabling stronger cuts may lead to faster convergence on harder problems.

For TS, SA, and SD algorithms, common hyperparameters include initial_states and num_reads. The initial_states parameter sets the number of distinct starting points used to initialize the solver, promoting diversity in the search and reducing sensitivity to poor initial guesses. The num_reads parameter determines how many times the solver independently attempts to find a solution; each run is performed from scratch, and the best result is retained. Increasing this value improves robustness and solution quality but scales linearly with computational time.

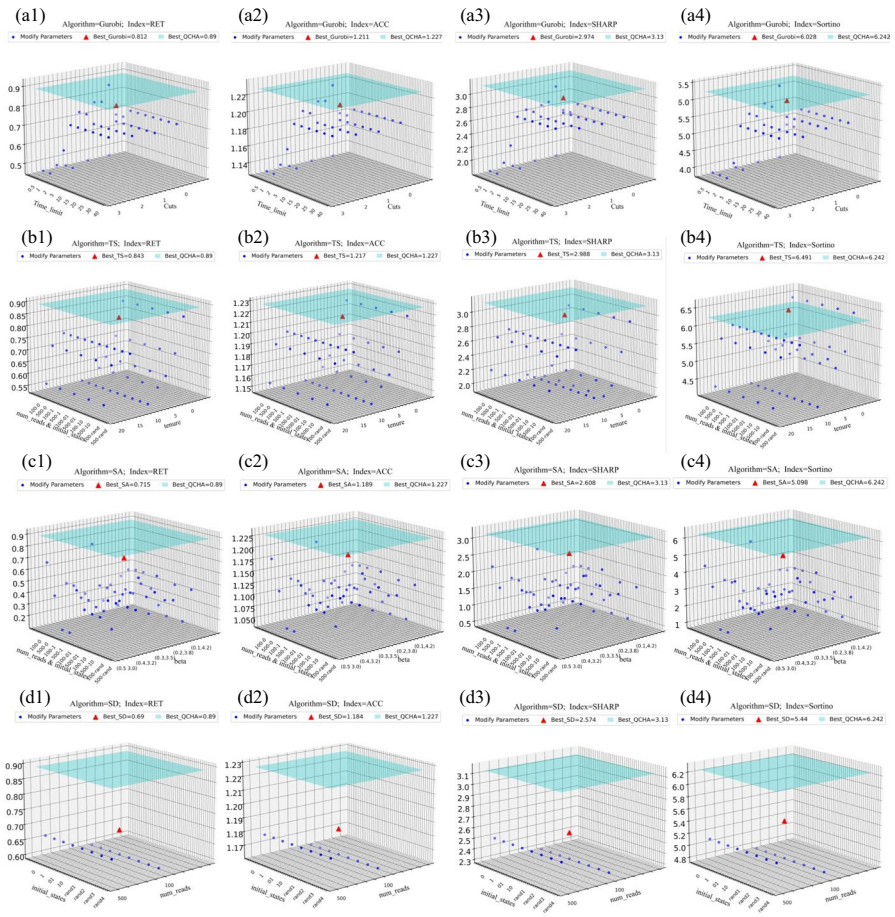

Fig. 6 Scatter plot illustrating the results of hyperparameter tuning for classical algorithms on Dataset-1. Blue points represent investment portfolio strategies and their corresponding back-testing metrics under different parameter configurations. The red triangle denotes the best-performing solution identified by the current classical algorithm. The light blue plane indicates the optimal results achieved by the QCHA algorithm. The X and Y axes represent tunable hyperparameters, while the Z axis shows the value of the back-testing metric. Numerical indices correspond to evaluation metrics: (1) RET, (2) ACC, (3) SHARPE, and (4) Sortino. Alphabetical indices denote classical solvers: (a) Gurobi, (b) TS, (c) SA, and (d) SD

Specifically for the SA algorithm, we additionally tune the annealing schedule by varying the inverse-temperature range, denoted as beta_range $= (\beta_{\min}, \beta_{\max})$. The lower bound $\beta_{\min}$ (corresponding to a high temperature) allows more frequent acceptance of worse solutions at the start, enabling global exploration, while the upper bound $\beta_{\max}$ (low temperature) enforces greedy descent toward a local minimum in later stages. The tested schedules include: (0.1, 4.2), (0.2, 3.8), (0.2, 3.8), (0.3, 3.5), (0.4, 3.2), and (0.5, 3.0). These schedules balance exploration and exploitation and help prevent premature convergence. A rule of thumb in tuning β_{range} is to begin

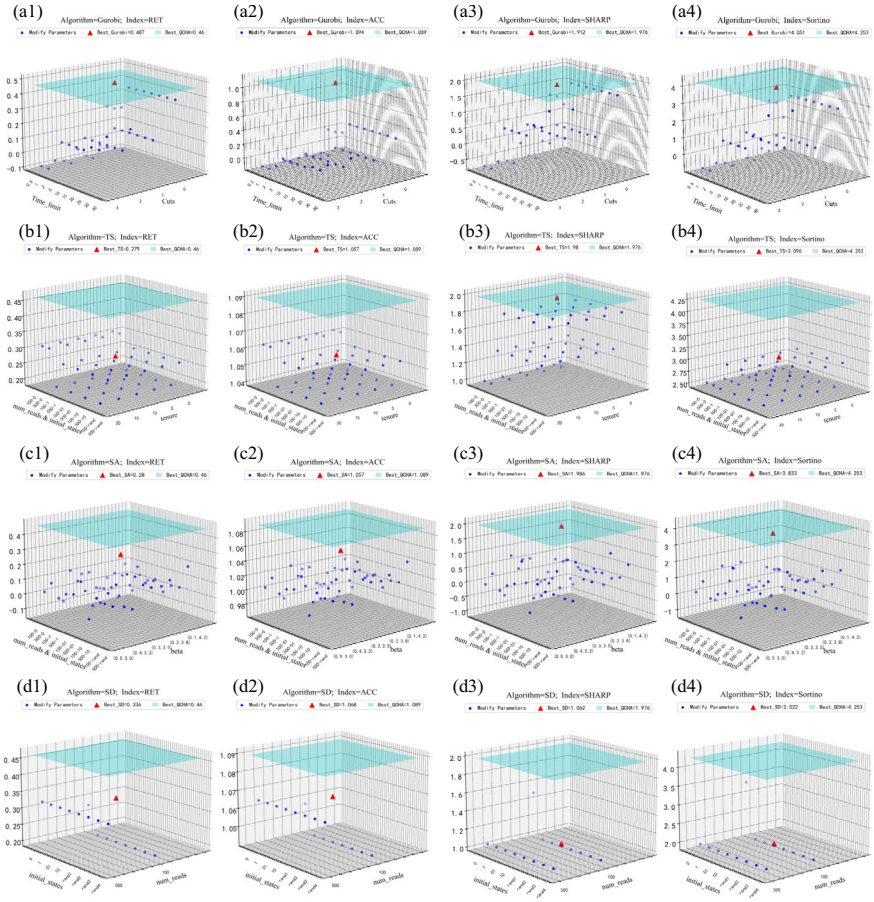

Fig. 7 Scatter figure by adjusting hyper-parameters of the classical algorithms on Dataset-2

with an initial acceptance rate around 80% and cool down until the acceptance rate drops below 1%.

D Supplementary Materials for Experimental Results

In this section, we present detailed experimental results, including the allocation proportions of the optimal strategies and the back-testing results. Additionally, we offer further analysis and discussions.

D.1 Using Six Algorithms to Solve PO Problem on Dataset-1

In this study, we present the results of portfolio optimization conducted using six distinct algorithms on Dataset-1. Here, we present the portfolio allocations resulting from each algorithm, as illustrated in Table 1. The corresponding back-test index

Table 1 The allocation proportion of the optimal portfolio strategies on Database-1

Algorithms	600017SH	300148SZ	002451SZ	601919SH	000539SZ	600830SH	600763SH	601008SH	600519SH	000858SZ
QCHA	0.0	0.097	0.0	0.0	0.258	0.032	0.258	0.323	0.0	0.032
Gurobi	0.0	0.129	0.0	0.0	0.258	0.0	0.194	0.29	0.0	0.129
QA	0.0	0.161	0.0	0.0	0.161	0.065	0.29	0.194	0.032	0.097
TS	0.0	0.065	0.0	0.0	0.225	0.0	0.258	0.323	0.032	0.097
SA	0.0	0.065	0.032	0.0	0.258	0.097	0.161	0.129	0.0	0.258
SD	0.065	0.097	0.0	0.0	0.194	0.032	0.194	0.225	0.0	0.193

These strategies are calculated by six different algorithms, such as QCHA, Gurobi, QA, TS, SA, and SD, when the total number of discrete independent variables used in constructing model is 50

Table 2 Performance of portfolio optimization given by six algorithms based on Dataset-1

Algorithms	Hamiltonian	RET	ACC	Sharpe	Sortino
QCHA	-9.999647587	0.890	1.227	3.130	6.242
Gurobi	-9.9997203	0.812	1.211	2.974	6.028
QA	-9.999394085	0.781	1.204	2.728	6.246
TS	-9.999600907	0.843	1.217	2.988	6.491
SA	-9.999503829	0.715	1.189	2.608	5.098
SD	-9.99957352	0.690	1.184	2.574	5.440

There are five columns in the table, and each column is a measurement index, including Hamiltonian, RET, ACC, Sharpe Ratio, and Sortino Ratio. Each row represents the numerical results calculated using six different algorithms, respectively

results for the schemes generated by the different algorithms are displayed in Table 2. According to Table 1, several poorly performing stocks, such as 600017SH and 002451SZ, were excluded from investment when applying the QCHA, Gurobi, and QA algorithms. In contrast, the other classical algorithms failed to identify and avoid these poorly performing stocks. The solution provided by the QCHA algorithm exhibits minor variations in allocation proportions compared to Gurobi's results. Consequently, the QCHA algorithm ultimately yields superior outcomes, underscoring the advantages of quantum computing, as evidenced in Table 2.

The QCHA algorithm outperforms all other algorithms across every metric. It achieves the highest results in three metrics: $RET = 0.890$, $ACC = 1.227$, and $Sharpe = 3.13$, suggesting that it is the most effective algorithm for portfolio optimization among those tested. The Sortino ratio provided by QCHA ranks second, slightly lower than that of the TS algorithm. The classical algorithms, TS and Gurobi, demonstrate moderate performance, which is slightly inferior to that of QCHA, indicating a less favorable risk-adjusted return. The remaining three algorithms exhibit significant disadvantages across these indicators. Regarding the QA algorithm implemented by D-Wave's quantum annealing machine, its performance surpasses that of the SD SA algorithm, though the margin is modest. In summary, the QCHA algorithm emerges as an effective method for portfolio optimization, outperforming two commonly used classical algorithms: TS and Gurobi. Additionally, QCHA is a significantly more robust hybrid algorithm compared to simple quantum annealing.

D.2 Using 5 Algorithms to Solve PO Problem in Dataset-2

The allocation ratios of the portfolio optimization strategy on Dataset-2 are presented in Table 3. A comparison reveals that the strategies generated by each algorithm are not entirely identical. Nevertheless, the top three performing algorithms, QCHA, TS, and Gurobi, exhibit a shared tendency to avoid investing in certain underperforming stocks. This observation suggests that our model possesses strong capabilities in addressing practical financial investment challenges.

According to Table 4, the Sortino ratio obtained by QCHA is the best. Moreover, the QCHA algorithm ranks second in terms of ACC value and Sharpe ratio, trailing the best by only 0.45% and 0.2%, respectively. Therefore, QCHA remains the best algorithm. Overall, Gurobi, QCHA, and TS are still significantly better than the others according to the results of backtesting, each with its strengths and weaknesses.

Table 3 The allocation proportions of the optimal portfolio strategies are derived from five different algorithms, utilizing a total of 128 discrete independent variables

Algorithms	6005	6885	6010	6033	6000	5010	1617	5160	5010	1595	XT1613	XT1519	XT1623	XT1410	XT1510
	19.SH	43.SH	22.SH	09.SH	18.SH	80.OF	25.OF	70.OF	58.OF	08.OF	496.XT	228.XT	529.XT	420.XT	216.XT
QCHA	0.016	0.348	0.021	0.001	0.0	0.044	0.454	0.016	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Gourbi	0.0	0.409	0.007	0.0	0.0	0.001	0.371	0.052	0.06	0.0	0.0	0.0	0.0	0.0	0.1
TS	0.0	0.0	0.0	0.0	0.0	0.063	0.36	0.034	0.228	0.015	0.1	0.0	0.0	0.1	0.1
SA	0.049	0.094	0.033	0.002	0.003	0.048	0.323	0.113	0.135	0.0	0.0	0.0	0.1	0.0	0.1
SD	0.033	0.464	0.049	0.014	0.013	0.004	0.108	0.108	0.001	0.006	0.1	0.0	0.0	0.1	0.0

This analysis is based on Dataset-2, which includes three types of assets with varying degrees of dispersion

Table 4 The performance indicators for portfolio optimization, calculated using five algorithms in Dataset-1, are summarized in the table

Algorithms	Hamiltonian	RET	ACC	Sharpe	Sortino
QCHA	-1000.017372	0.460	1.089	1.976	4.253
Gurobi	-1000.01746	0.487	1.094	1.912	4.051
TS	-1000.016794	0.279	1.057	1.980	3.096
SA	-1000.016798	0.280	1.057	1.986	3.833
SD	-1000.017077	0.336	1.068	1.062	2.022

This table consists of five columns, each representing a measurement index: Hamiltonian, RET, ACC, Sharpe Ratio, and Sortino Ratio. Each row displays the numerical results obtained from the five different algorithms

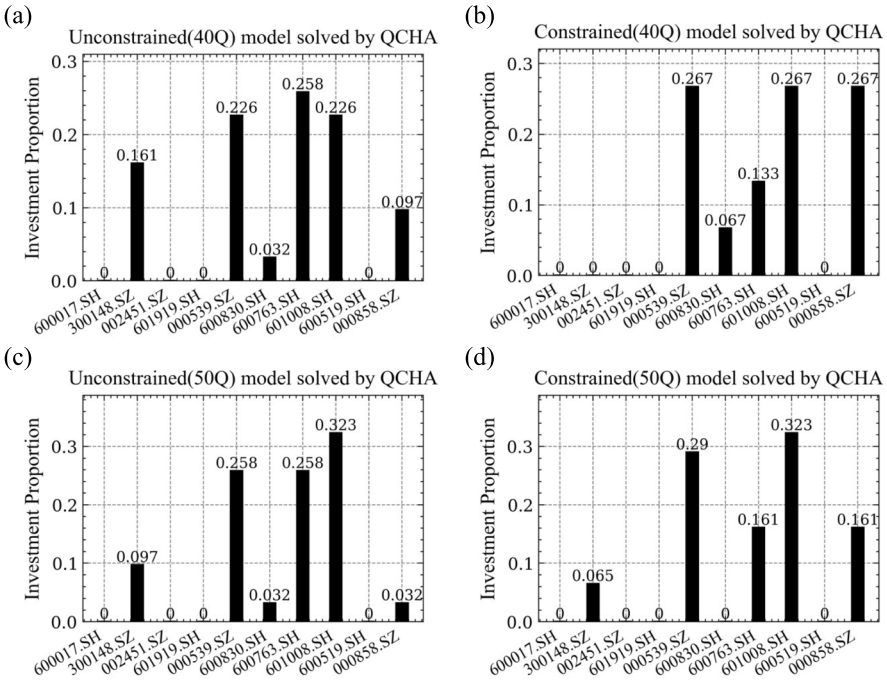

Fig. 8 The allocation ratios of the investment strategy following portfolio optimization using the QCHA algorithm are presented in this section. The experiments were conducted on Database-1, where the stock codes of ten assets are labeled on the x-axis. The number of independent variables is set at (a, b) $N = 40$ and (c, d)$N = 50$, respectively. Information regarding whether constraints are considered in the model is indicated in the titles above each figure

In this case, due to the number of independent variables exceeding hardware limitations, traditional quantum annealing algorithms cannot be executed. In summary, QCHA is a very effective method for solving portfolio optimization problems.

D.3 Results of Portfolio Optimization with and Without Constraints

In the main paper, we compared the backtesting results of various algorithms under two scenarios: considering and not considering constraints, with the number of independent variables set at 40 and 50, respectively. Due to the reduced solution space and variations in boundary conditions, all evaluation indicators show a decrease of approximately 10% compared to the results without constraints. Figure 8 illustrates the optimal investment strategies provided by the QCHA algorithm under different conditions.

As shown in Fig. 8, the number of stocks available for investment decreases, while the concentration of investments increases after considering the constraints. Additionally, as the number of qubits increases, the changes in the allocation pattern of investment proportions become more subtle when comparing allocations with different numbers of qubits used. This finding demonstrates that the QCHA algorithm possesses strong capabilities. Assuming we have obtained an optimal solution without considering constraints, introducing additional constraints and relaxation variables does not yield significant differences from it. Typically, the introduction of extra constraints leads to a deterioration in the optimal solution as the number of qubits increases, particularly because the problem size grows exponentially with the increase in qubits. Therefore, it is significant that the QCHA algorithm can still maintain equivalent capabilities in solving portfolio optimization problems. In summary, QCHA outperforms other classical algorithms.

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Author Contributions Haijing Wei, Wending Zhao, and Lin Yao conceived the experiments and theory. Yanbo J. Wang and Qi Xu conducted the experiments. Haijing Wei and Wending Zhao analysed the results. Wending Zhao, and Lin Yao wrote this manuscript. All authors reviewed the manuscript.

Data and code availability All of the data are available upon request. Source codes covered in this paper can be made available upon request.

Declarations

Competing interests All of the authors declare on behalf of all authors that there are no competing interests.

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